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Research article

Offspring sex ratio in a communal breeding bird is male-biased when pre-breeding rainfall is low

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Offspring sex ratios may deviate from parity when the fitness benefits of producing male or female offspring vary. We tested for sex ratio bias in smooth-billed anis *Crotophaga ani*, a communal laying cuckoo with low within-group relatedness and high offspring dispersal. One male group member performs nocturnal incubation and sires more offspring than other males in the group, suggesting males may have greater reproductive variance than females. We hypothesized that pre-breeding rainfall influences food availability and offspring sex ratio, predicting that breeding females skew production towards the sex with higher reproductive variance (males) in high food years. Females may also adjust sex ratio across the hatching order to increase survival of the more competitive sex, especially when clutches are larger and within-brood competition is higher. As adults, male smooth-billed anis are larger than females, so we assumed male nestlings are more competitive than females and predicted a male-bias in first hatched chicks in larger broods. Contrary to our first prediction, offspring sex ratio was male biased when pre-breeding rainfall was lower. In partial support of our second prediction, marginally more first hatched chicks were male in larger broods. To our knowledge, this is the first evidence of offspring sex ratio bias in a communal laying bird species. Future work in this system will attempt to uncover the mechanisms by which co-breeding females adjust offspring sex ratio and test alternative hypotheses to explain male-biased offspring sex ratios under different conditions.

Keywords: communal laying, cooperative breeding, *Crotophaga ani*, joint laying, male bias, offspring sex ratio

Introduction

Sex ratio analyses have often been guided by two major hypotheses proposed by Fisher (1930) and Trivers and Willard (1973). Fisher postulated that parents should invest resources equally in male and female offspring, since each sex contributes half the



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